



EXCRETORY PRODUCTS AND THEIR ELIMINATION

1 EXCRETORY WASTE

METABOLISM **EXCESS INGESTION**

Results in accumulation of

- **Nitrogenous wastes** – NH_3 , Urea, Uric acid
- Other contents CO_2 , H_2O , ions (Na^+ , K^+ , Cl^- , PO_4^{3-} , SO_4^{2-})

Removed Partially/Completely

3 EXCRETORY STRUCTURES

- Most invertebrates – Simple tubular forms
- Vertebrates – Complex tubular organs called kidneys

Structures

- Protonephridia/flame cells
- Nephridia
- Malpighian tubules
- Antennal/Green glands

Examples

- Platyhelminthes (Planaria)
- Rotifers
- Some annelids
- Cephalochordates (Amphioxus)
- Annelids (Earthworms)
- Insects (Cockroaches)
- Crustaceans (Prawn)

- **Protonephridia** are primarily concerned with **osmoregulation**
- **Functions of excretory structures**
- Eliminate nitrogenous wastes
- Maintain ionic and acid-base balance of body fluids. i.e., osmoregulation

MICTURITION

- Process of release of urine
- Mechanism – **Micturition reflex**

Urinary bladder (Store Urine)
Activates

Stretch receptors

signals

CNS (voluntary signal)

↓ Send motor message

Urinary bladder

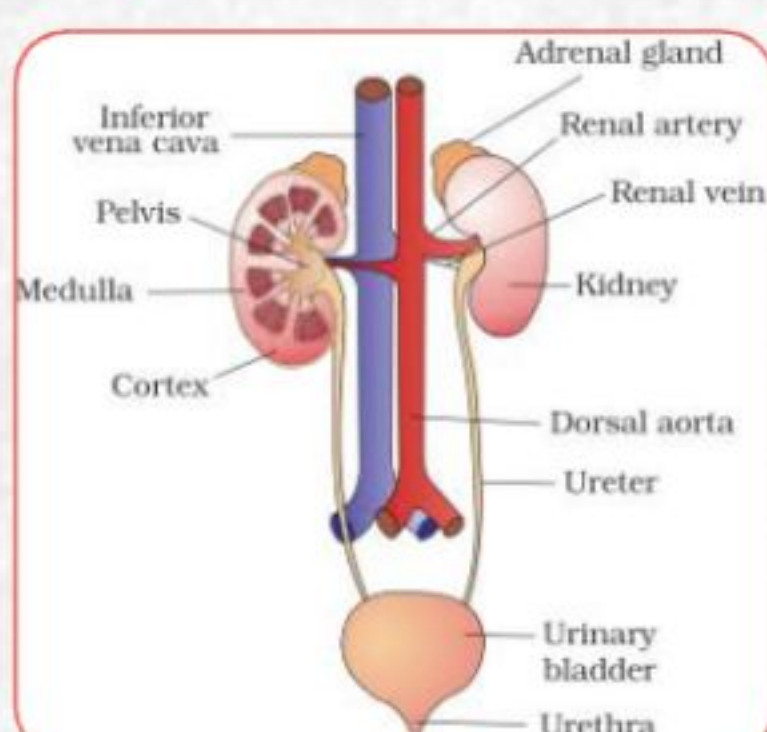
• Smooth muscle contract

• Urethral sphincters relax

↓ Release

Urine

4 HUMAN EXCRETORY SYSTEM



KIDNEY

- 1 pair, bean shaped, reddish brown.
- Length 10–12 cm, Width 5–7 cm Thickness 2–3 cm.
- Weight 120–470 g.
- Between T_{12} – L_3 vertebra, close to dorsal inner wall of abdominal cavity.

URINARY BLADDER

- Have stretch receptors.
- Store urine till voluntary signals from CNS carries out its release.

URETHRA

- Guarded by sphincters.
- Meant for release of urine.

2 NITROGENOUS WASTES

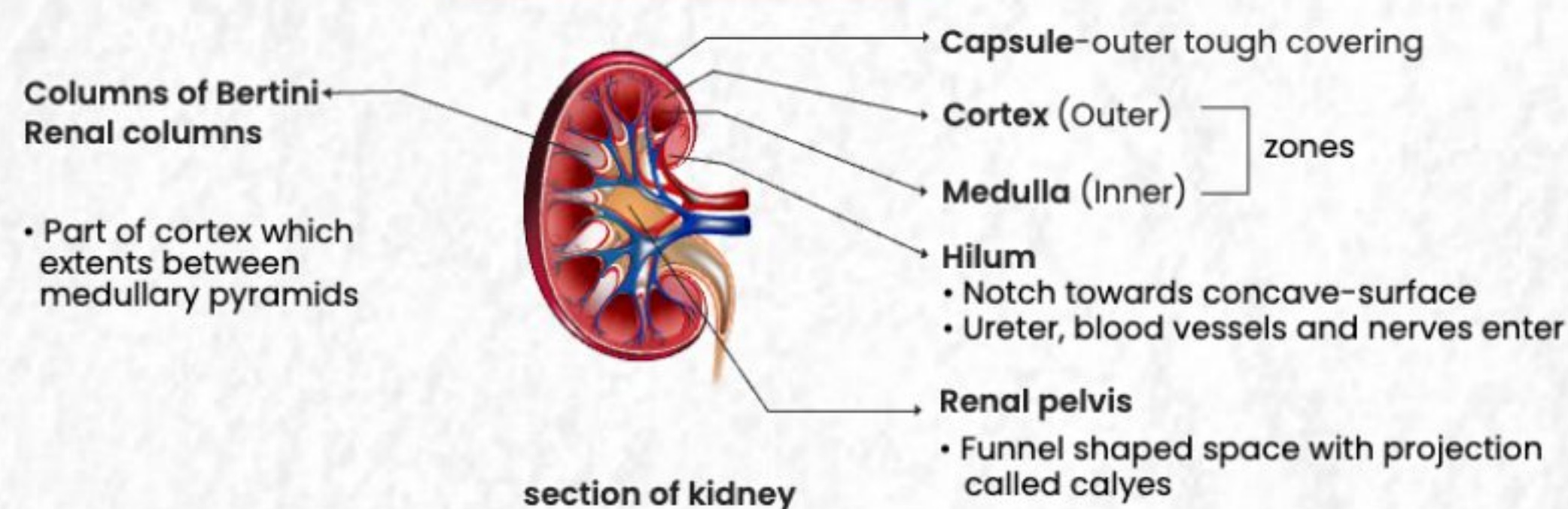
- Nature of nitrogenous waste formed and their excretion vary among animals depending on the **habitat/availability of water**

Major Nitrogenous Waste	Nature & Examples	Toxicity And Water Required	Typical
• Ammonia	Ammonotelic	Maximum	• Diffusion through gills surface or body surface ammonium ions
• Urea	Ureotelic	Less	• Kidneys filter urea from blood
• Uric acid	Ureotelic • Land snads • Insects • Reptiles • Birds	Least	Pellet/Paste

• Ammonia converts into urea in liver via ornithine cycle.
 • Elimination of urea, uric acid is meant for conservation of water i.e., a type of terrestrial adaptation.
 • Kidneys do not play a significant role in removal of ammonia.

• Some amount of urea may be retained in the kidney
 • matrix of some animals to maintain desired osmolarity

5 KIDNEY

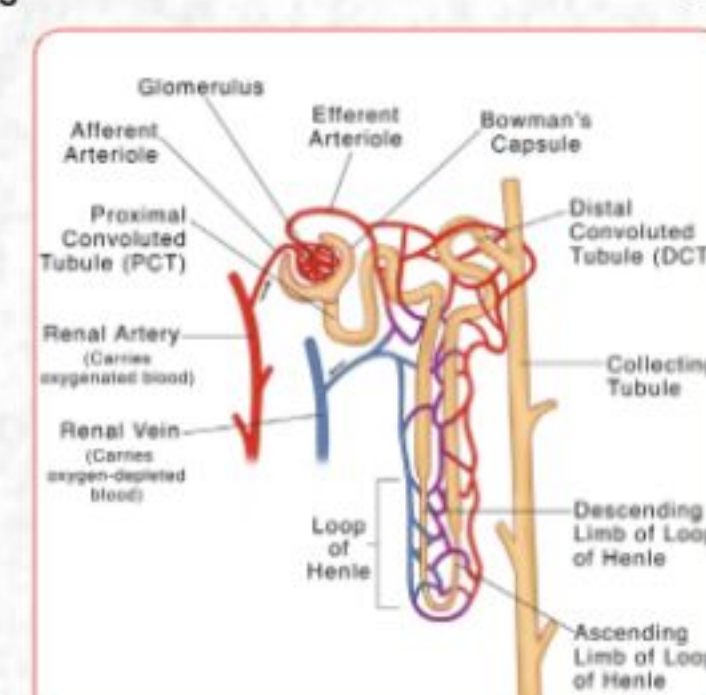


Medullary pyramids are conical masses that project into calyces.

6 NEPHRON

- **Functional unit of kidney**
- Each nephron has two parts

- Nearly **1 million/kidney**
- Parts: (i) Glomerulus (ii) Renal tubule





Parameters	Cortical	Juxtamedullary
• Number	More	Less
• Loop of Henle	Too short	Very long
• Extension Into medulla	Very Mild	Deep
• Vasa recta	Absent/reduced	Present

• Juxtaglomerular apparatus (JGA)

Sensitive region formed by cellular modifications in **distal convoluted tubule** and **afferent arteriole** at the location of their **contact**

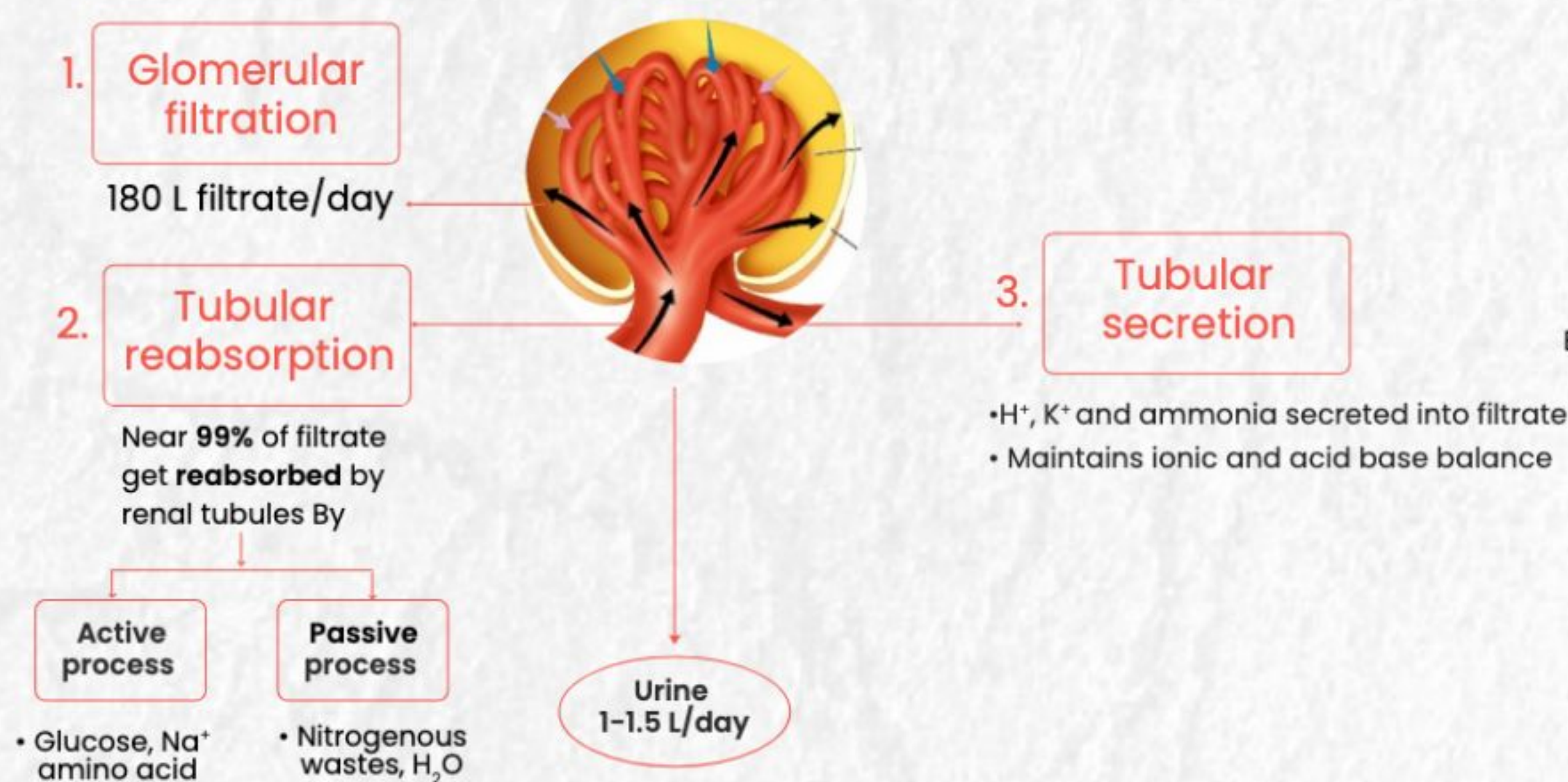
- Nephrons are dipped in interstitial fluid having specific osmolarity
- Cortex - 300 mOsm/L - Medulla - upto 1200 mOsm/L (Gradient)

- **Glomerulus** is a tuft of capillaries formed by Afferent Arteriole. A fine branch of Renal Artery.
- **Malpighian Corpuscle, PCT, DCT – Located In Cortex.**
- Loop of Henle – Dips into Medulla.
- Many Ducts open into straight tube called collecting duct, many of which converge into renal pelvis through mediullary Pyramids in the calyces.
- **Efferent Arteriole** emerging from glomerulus forms peritubular capillaries around renal tubule
- **Vesa recta:**
Branch of petitubular capillaries
Parallel to loop of Henle
'U' shaped

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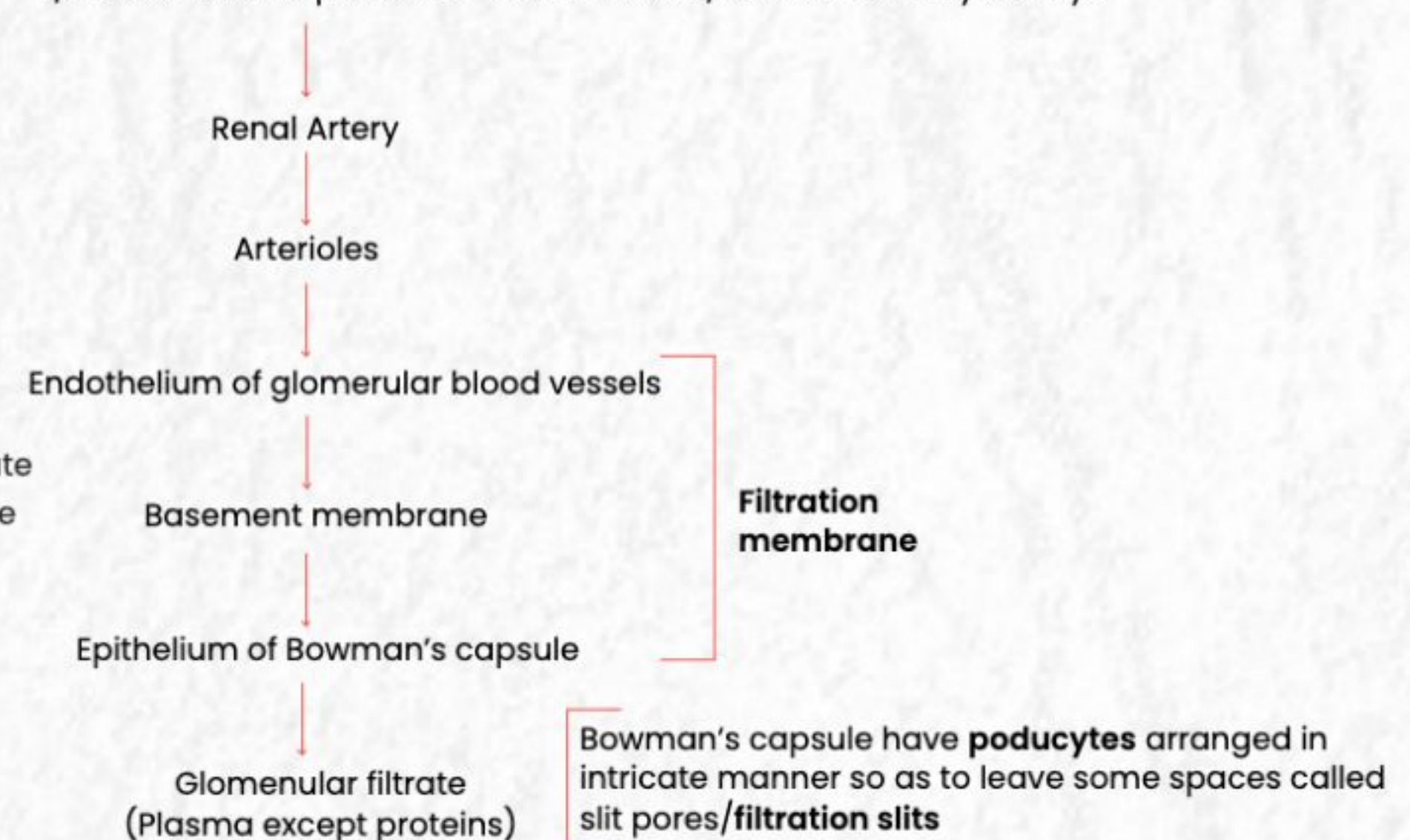
URINE FORMATION

Main Processes



Glomerular Filtration/Ultra filtration

-1/5 of cardiac output or 1100-1200 ml blood/min is filtered by kidneys

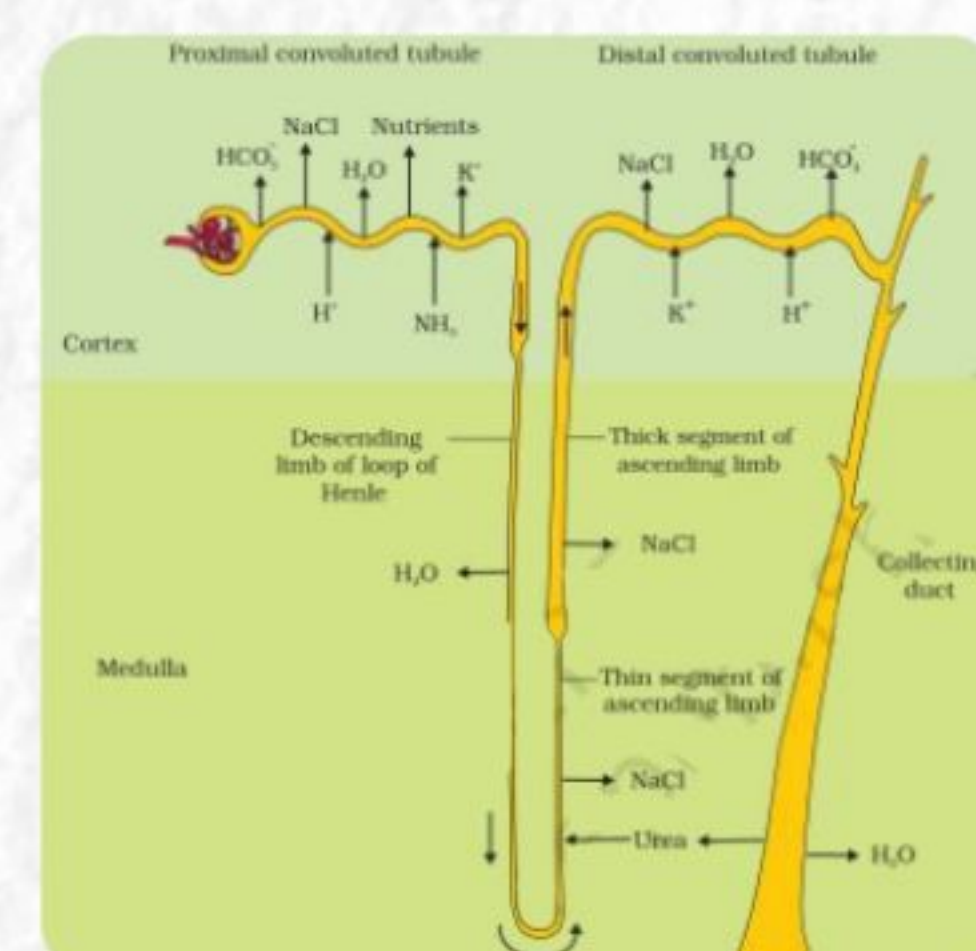


• Filtration is due to pressure in the glomerular capillaries.

• **Glomerular filtration rate (GFR) = Filtration/min 125ml/min**

• Kidney has an ability to regulate GFR.

- Lined by simple cuboidal brush border epithelium
- Nearly all essential nutrients, 70-80% electrolytes and water are reabsorbed
- Major site of reabsorption & selective secretion



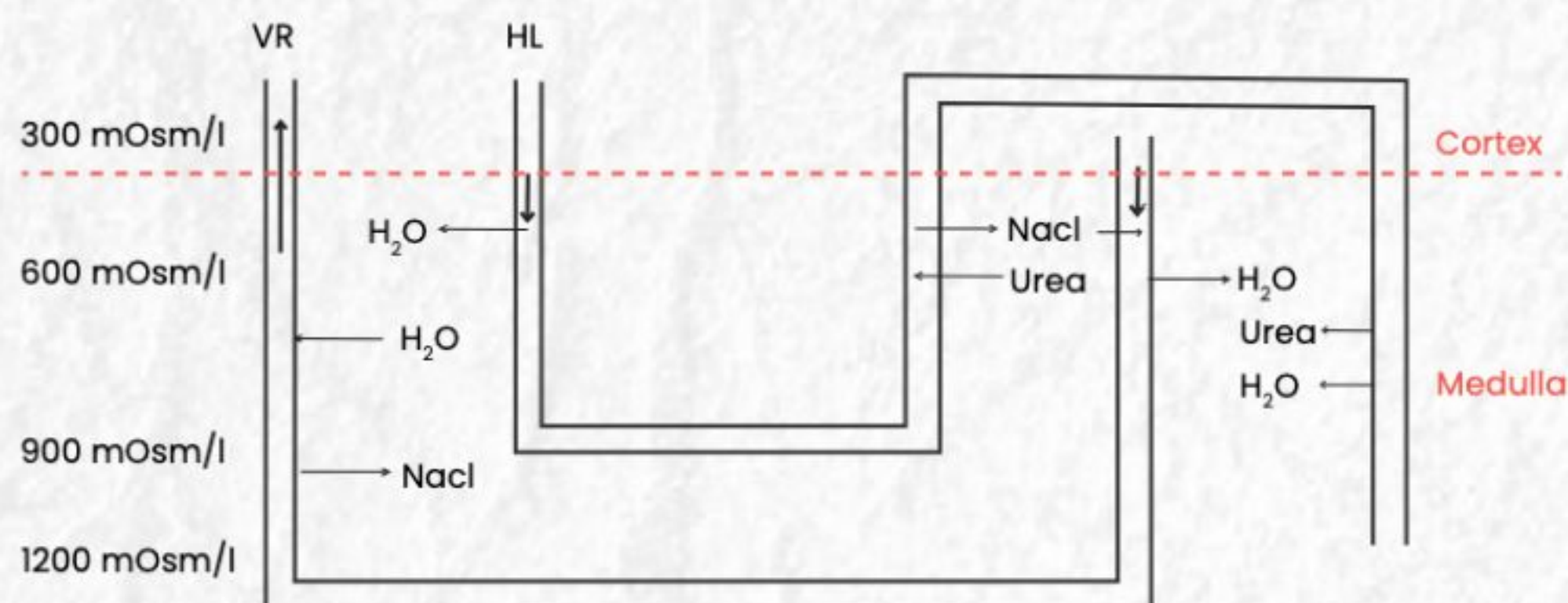


8 COUNTER CURRENT MECHANISM TO CONCEITTRATE FILTRATE

• Flow of filtrate in different limbs of following structures are in opposite direction (Counter current)

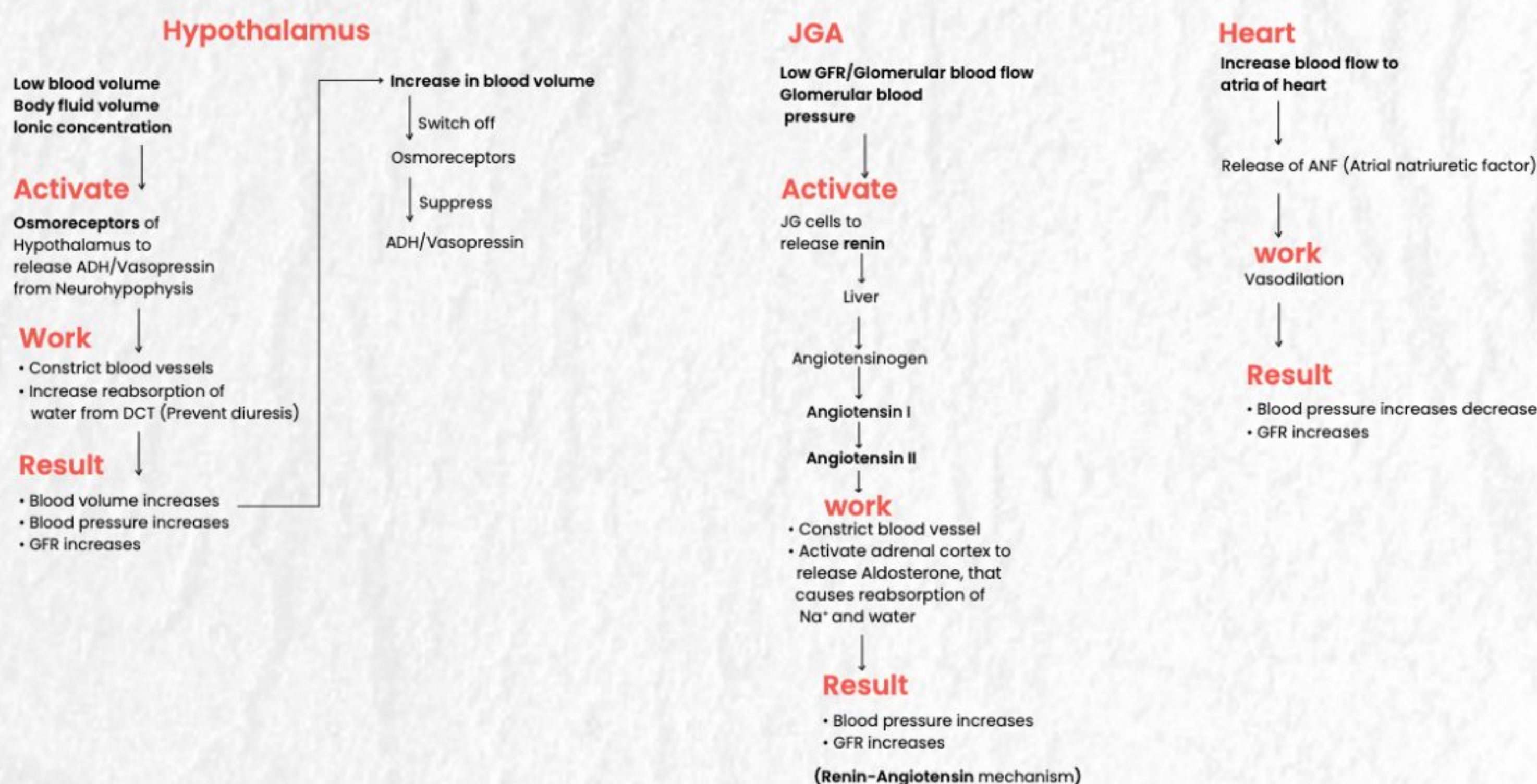
• Loop of Henle (HL) • Vasa Recta (VR) • HL and VR

- Proximity of HL and VR and counter current in them increase osmolarity towards inner medullary interstitium (300 mOsm/L in cortex to 1200 mOsm/L)
- Interstitium gradient is caused by NaCl and urea.
- NaCl transported by ascending limb of HL exchanged with descending limb of vasa recta and is returned to medullary interstitium by ascending limb of VR.
- Urea which enters in thin part of ascending limb of HL is transported back to interstitium by collecting tubule
- This mechanism maintain interstitial concentration gradient that helps in easy passage of water from collecting tubule thereby concentrating filtrate (urine).



- Mammals have ability to poroduce concentrated urine.
- HL primarily helps to maintain osmolarity gradient in kidney intetstitium.

9 REGULATION OF KIDNEY FUNCTION/GLOMERULAR FILTRATION RATE



ANF mechanism acts as a check on Renin-Angiotensin mechanism

10 CHARACTERISTICS AND COMPOSITION OF URINE

Colour - Light yellow

pH = 6

Odour - Characteristic

Human kidneys can produce urine nearly **4 times** concentrated than initial filtrate.

Urea - 25-30 gm/day

Various conditions can affect characteristics of urine.

Abnormal constituents of urine	Condition	Indicate
Glucose	Glucosuria	Diabetes mellitus
Ketone bodies	Ketonuria	Diabetes mellitus



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DISORDERS OF EXCRETORY SYSTEM

Disorders	Symptoms or Treatment
Renal catculi	Stone or insoluble mass of crystalised salts (e.g. oxalates)
Glomerulonephritis	Inflammation of glomeruli of kidney
Renal/kidney failure	Malfunctioning of kidneys leads to accumulation of urea in blood (Uremia); highly harmful, may lead to kidney failure. Treatment (i) Haemodialysis: Process to remove urea from blood Method is a boon for thousands of uremic patients all over the world. <pre> graph LR A[Blood drained from artery] -- Mix with --> B[Heparin Anticoagulant] B -- Pumped through --> C[Dialysing unit] subgraph Dialysing_unit [Dialysing unit] D[Artificial kidney] E[Porous cellophane tubes surrounded by dialysing fluid.] end C -- Filtration based on concentration gradient --> F[Nitrogenous wastes freely move out] F --> G[Clear blood] G -- Mix with --> H[Anticoagulant] H -- Pumped back to body through Vein --> I[] </pre> (ii) Kidney transplantation Ultimate method in correction of acute renal failure - Functional kidney is taken from donor - To minimize rejection, close relatives are preferred as donor - Modern clinical problems have increased success rate of such complicated techniques

Composition of dialysis fluid is same as plasma except the nitrogenous wastes

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ROLE OF OTHER ORGANS IN EXCRETION

Accessory structure	substances	Basic Work
Lungs	CO ₂ , water	- Remove large amount of CO₂, approximately 200 mL/min - Remove significant quantity of water
Liver (Largest gland)	Bilirubin, vitamins biliverdin, drugs cholesterol, degraded steroid hormones	Most of these substances pass out alongwith digestive wastes
Skin • Sweat glands	Sweat contains • NaCl • Urea • Lactic acid	Primary function of sweat is to facilitate cooling effect on body surface
• Sebaceous glands	Sebum contains • Sterols • Hydrocarbons • Waxes	Sebum provides a protective oily covering for the skin
Salivary glands		Small amount of nitrogenous wastes are eliminated through saliva